

Permutation test

Test a difference by shuffling the group labels.

1. Start

Take the observed data. Compute the observed statistic.

Group A = {2, 4, 6} mean = 4

Group B = {5, 7, 9} mean = 7

diff = 3

2. Shuffle the labels — no replacement

Pool all six values. Give three back to each group. Keep the sizes equal.

Pooled



One shuffle



A*



B*

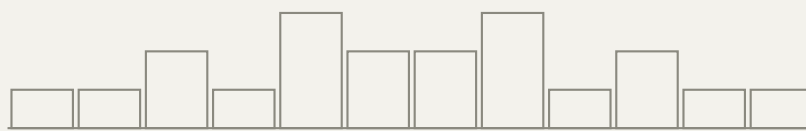
3. Recompute the statistic

Use the shuffled groups. Compute the statistic again.

$$\text{mean}(B^*) - \text{mean}(A^*) = 6 - 5 = 1$$

4. Build the null distribution

Repeat the shuffle B times. Here list all 20 splits for the exact test.



centre = 0

5. Read the p-value

Mark the observed value. Shade the values as far from 0 as the observed value.



$$p = 4 / 20 = 0.20$$

repeat B times

Why this works

If the groups are the same, the labels carry no information.

So a label shuffle shows the statistic under the null.

Assumption: exchangeability under the null.

 Group A

 Group B

 shaded bars = as far from 0 as the observed value

Worked example: two groups of three. Statistic = $\text{mean}(B) - \text{mean}(A)$.